

Understanding Cloud Computing-34

Introduction

Cloud computing is the delivery of computing services over the internet. Instead of running applications or storing data on local servers or personal computers, cloud computing provides access to servers, storage, databases, networking, and software hosted in remote data centers. Users pay only for what they use, which makes cloud computing cost-efficient, scalable, and flexible.



Key Characteristics of Cloud Computing

1. **On-demand self-service** – Users can provision computing resources automatically without human intervention.
2. **Broad network access** – Services are accessible over the internet from desktops, laptops, and mobile devices.
3. **Resource pooling** – Multiple customers share the same infrastructure, but their data remains isolated.
4. **Rapid elasticity** – Resources can scale up or down automatically depending on demand.
5. **Measured service** – Cloud providers measure resource usage and charge based on consumption.

8 Key Features of Cloud Computing



Cloud Service Models

Cloud services are offered in three main models:

1. Infrastructure as a Service (IaaS)

- Provides virtualized computing resources like servers, storage, and networking.
- Users install and manage their own operating systems and applications.
- Example: Amazon EC2, Microsoft Azure VMs, Google Compute Engine.

2. Platform as a Service (PaaS)

- Provides a ready-to-use platform with operating systems, databases, and development tools.
- Developers only focus on building applications, not managing infrastructure.
- Example: Google App Engine, AWS Elastic Beanstalk, Microsoft Azure App Services.

3. Software as a Service (SaaS)

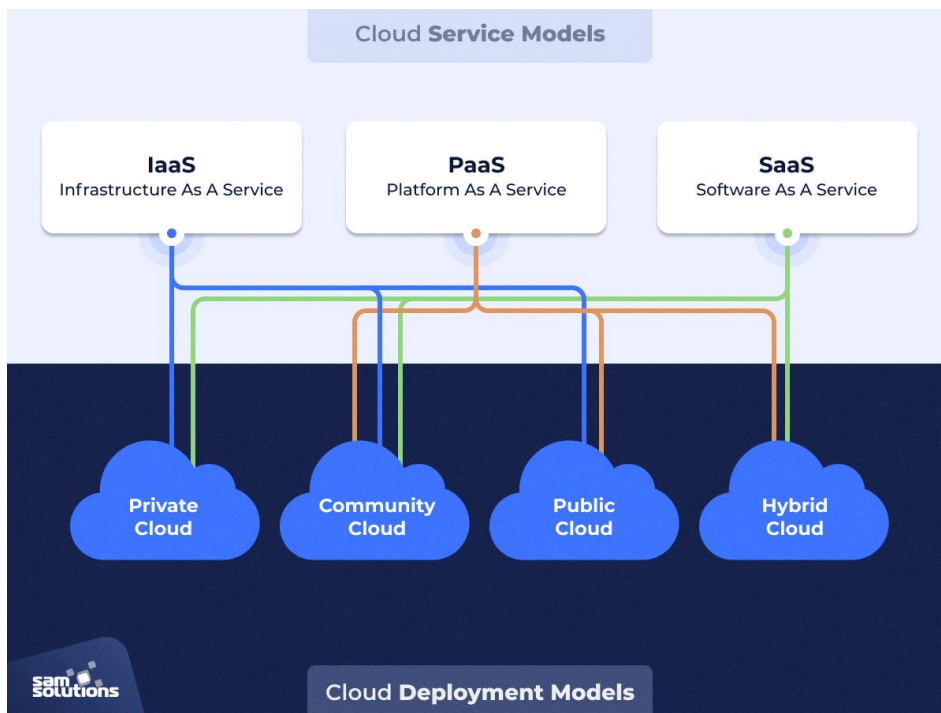
- Provides complete applications delivered over the internet.
- Users access them via browsers without worrying about installation or maintenance.
- Example: Gmail, Office 365, Salesforce, Dropbox.

Types of Cloud Computing Services



Cloud Deployment Models

1. **Public Cloud** – Services are provided over the public internet by vendors like AWS, Azure, or Google Cloud.
2. **Private Cloud** – Infrastructure is dedicated to a single organization for enhanced security and control.
3. **Hybrid Cloud** – Combination of public and private clouds, allowing workload mobility between them.
4. **Community Cloud** – Shared infrastructure for organizations with similar requirements (like government agencies or healthcare).



Advantages of Cloud Computing

- **Cost savings** – No need to buy and maintain expensive hardware.
- **Scalability** – Expand or reduce resources based on demand.
- **High availability** – Cloud providers guarantee uptime with multiple data centers.
- **Security** – Providers offer encryption, access controls, and compliance standards.
- **Global access** – Services are accessible from anywhere with internet connectivity.

Challenges of Cloud Computing

- **Downtime** – Cloud outages can impact business operations.
- **Data security** – Data stored in third-party data centers may face privacy concerns.
- **Vendor lock-in** – Switching providers can be difficult due to differences in platforms.
- **Compliance** – Organizations must meet regulatory standards like GDPR or HIPAA.

Real-World Use Cases

1. **Business Applications** – Gmail, Microsoft Teams, Zoom.
2. **Storage and Backup** – Google Drive, Dropbox, AWS S3.
3. **Big Data Analytics** – Cloud-based tools analyze massive datasets (e.g., AWS Redshift, Google BigQuery).
4. **Artificial Intelligence and Machine Learning** – Cloud AI platforms provide ready-made ML models (e.g., Azure AI, AWS SageMaker).
5. **Virtual Desktops** – Access to remote desktops from anywhere using cloud VDI.

Popular Cloud Providers

1. **Amazon Web Services (AWS)** – Largest market share, provides IaaS, PaaS, SaaS.

2. **Microsoft Azure** – Strong in hybrid cloud and enterprise integrations.
3. **Google Cloud Platform (GCP)** – Strong in AI, analytics, and Kubernetes.
4. **IBM Cloud** – Focus on enterprise hybrid solutions.
5. **Oracle Cloud** – Popular for database-related services.



Importance in Modern IT

- Businesses shift to cloud to reduce costs and increase agility.
- Startups use cloud to scale without large upfront investments.
- Governments adopt cloud for citizen services.
- Cloud supports remote work and global collaboration.
- AI, IoT, and big data heavily depend on cloud infrastructure.